

92. Lookahead aur Lookbehind

Kabhi kabhi sirf wahi matches chahiye jinke **baad** ya **pehle** koi aur pattern ho. Iske liye "lookahead" aur "lookbehind" (milkar "lookaround"). Example: `(1 turkey costs 30€)` se price (number jiske baad `€` ho).

Lookahead: `X(?:Y)`

"`X` dhundho, par tabhi match karo jab uske **baad** `Y` ho." `X` aur `Y` koi bhi pattern.

```
js
"1 turkey costs 30€".match(/\d+(?=€)/); // 30 (1 ignore, kyunki uske baad €
```

Lookahead sirf ek **test** hai. `(?=...)` ke andar ka content result (`30`) me shamil **nahi** hota.

Engine `X` dhundhta hai, phir dekhta hai ki turant baad `Y` hai ya nahi; nahi to potential match skip.

Kai tests: `X(?:Y)(?=Z)` = `X` ke baad `Y` aur `Z` dono ek saath (ye tabhi possible jab `Y`, `Z` aapas me exclusive na hon). Jaise `\d+(?=\s)(?=.*30)`: digits jinke baad space ho aur aage kahin `30` bhi ho:

```
js
"1 turkey costs 30€".match(/\d+(?=\s)(?=.*30)/); // 1
```

Negative lookahead: `X(?:!Y)`

"`X` dhundho, par tabhi jab uske baad `Y` **na** ho." Quantity chahiye (price nahi):

```
js
"2 turkeys cost 60€".match(/\d+\b(?:!€)/g); // 2 (price nahi mila)
```

Lookbehind

Lookbehind non-V8 browsers (Safari, IE) me supported nahi tha.

Lookahead "kya **baad** me aata hai" ki condition deta hai. Lookbehind "kya **pehle** hai" ki.

- **Positive lookbehind:** `(?<=Y)X`: `X` tabhi jab uske **pehle** `Y` ho.
- **Negative lookbehind:** `(?<!Y)X`: `X` tabhi jab uske **pehle** `Y` **na** ho.

Dollar amount (`$`) number ke **pehle**: `(?<=\$)\d+`

```
js
```

```
"1 turkey costs $30".match(/(?<=\$)\d+/); // 30
```

Quantity (number jiske pehle (\$) na ho): `(?<!\$)\d+`

```
js
```

```
"2 turkeys cost $60".match(/(?<!\$)\b\d+/g); // 2
```

Capturing groups

Lookaround ke parentheses ka content aam taur par result me nahi aata (jaise `\d+(?=€)` me `€` nahi aata). Par kabhi kabhi use (ya uske hisse ko) bhi capture karna hota hai: **extra parentheses** lagao.

```
js
```

```
"1 turkey costs 30€".match(/\d+(?=(€|kr))/); // 30, € (extra parentheses)
"1 turkey costs $30".match(/(?<=(\$|£))\d+/); // 30, $
```

Summary

Lookaround tab kaam aata hai jab kisi cheez ko uske context (pehle/baad) ke hisaab se match karna ho. Simple cases me khud bhi kar sakte hain (sab match karke loop me context filter), par lookaround aam taur par zyaada convenient hai.

Pattern	Type	Matches
<code>X(?=Y)</code>	Positive lookahead	<code>X</code> agar uske baad <code>Y</code>
<code>X(?!Y)</code>	Negative lookahead	<code>X</code> agar uske baad <code>Y</code> nahi
<code>(?<=Y)X</code>	Positive lookbehind	<code>X</code> agar <code>Y</code> ke baad
<code>(?<!Y)X</code>	Negative lookbehind	<code>X</code> agar <code>Y</code> ke baad nahi

Tasks

Non-negative integers (`0 12 -5 123 -18`) me se (`0, 12, 123`): `(?<![ -]\d+)` se `-18` ka `8` bhi mil jaata hai. Fix: `/((?![ -\d])\d+)/g` (pehle na `-` ho na koi digit, taaki number beech se shuru na ho).

`<body>` tag ke turant baad `<h1>Hello</h1>` insert karna (body me attributes ho sakte hain):

js

```
str.replace(</body.*?>/, '$&<h1>Hello</h1>'); // $& = match khud  
str.replace(/(?<=<body.*?>)/, `<h1>Hello</h1>`); // lookbehind: khaali stri
```

(s) aur (i) flags bhi kaam aa sakte hain: newline aur <BODY> ke liye.)